

The empirical recurrence for OEIS A251266: recovering a conjecture that is not written down

Adrian Perez Fontelles
Independent researcher

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Abstract

OEIS A251266 records “Empirical recurrence of order 62”, with the recurrence itself in a linked file rather than in the entry. The conjecture can nevertheless be settled, and the recurrence recovered, without reading that file. The entry counts arrays under a condition local to a bounded window of lines, so the count is a walk count on $S = 128$ vertices and satisfies some monic recurrence of order at most S ; Berlekamp–Massey applied to $2S$ exact terms therefore returns the MINIMAL such recurrence. Its order is exactly 62, the order the entry states, and the same is true of every tail of the sequence. Any recurrence of order 62 that the sequence satisfies from any point on must then have a characteristic polynomial that is a multiple of the minimal one and of the same degree, hence equal to it. So the entry’s recurrence is the one displayed here, whatever its file contains, and it is proved.

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1 A conjecture one cannot read

OEIS A251266 is “Number of $(n+1) \times (6+1)$ 0..1 arrays with no 2×2 subblock having $x_{11}-x_{00}$ less than $x_{10}-x_{01}$.”. It has offset 1 and begins

502, 3894, 31936, 279718, 2510004, 22806904, 208452452, 1910905110, ...

The entry states:

Empirical recurrence of order 62 (see link above)

As of the “Last modified” line on the live entry (Jun 02 2025 10:19:57, revision 6) this is still recorded as empirical. The recurrence’s coefficients are not in the entry text.

2 The count is a walk count

The entry’s condition is local: it constrains a bounded window of consecutive lines of the array and nothing further. The admissible configurations of one such window are the vertices of a finite digraph, an edge joining u to v when v may follow u , and an admissible array is exactly a walk. Hence, with M the adjacency matrix on $S = 128$ vertices, $a(n) = \iota^\top M^{j(n)} \tau$ for a linear reindexing j fixed by the entry’s shape and checked against its published terms. By the Cayley–Hamilton theorem M^S is an integer combination of $I, M, \dots, M^{S-1}$, so:

Lemma 1. *a satisfies a monic linear recurrence with integer coefficients of order at most $S = 128$.*

3 Recovering the recurrence

Lemma 2. *A sequence known to satisfy some linear recurrence of order at most S is determined, together with its minimal recurrence, by its first $2S$ terms: Berlekamp–Massey applied to them returns that minimal recurrence exactly.*

Running it on $2S = 256$ exact terms of the model returns order 62 — the order the entry states — with the integer coefficients

$$a(n) = \sum_{i=1}^{62} c_i a(n-i),$$

c_1	17	c_{17}	−45821756239	c_{33}	−9008260237392	c_{49}	681261808
c_2	−4	c_{18}	−81652713998	c_{34}	4820300303386	c_{50}	919620559
c_3	−1289	c_{19}	212175752502	c_{35}	6226822760871	c_{51}	−152283917
c_4	4559	c_{20}	218477930233	c_{36}	−4003964462680	c_{52}	−85288866
c_5	38517	c_{21}	−739431624692	c_{37}	−3398895252515	c_{53}	19759328
c_6	−215168	c_{22}	−414371492495	c_{38}	2591826477510	c_{54}	5156184
c_7	−561056	c_{23}	1993155643614	c_{39}	1439115269142	c_{55}	−1609299
c_8	4966817	c_{24}	481146175867	c_{40}	−1316423309481	c_{56}	−171385
c_9	3116763	c_{25}	−4226250353914	c_{41}	−457061226325	c_{57}	80119
c_{10}	−70412413	c_{26}	−44837732114	c_{42}	524132847357	c_{58}	1205
c_{11}	27376007	c_{27}	7124578240335	c_{43}	101058322790	c_{59}	−2198
c_{12}	672646288	c_{28}	−1177996515767	c_{44}	−162476312477	c_{60}	94
c_{13}	−703151511	c_{29}	−9608386181106	c_{45}	−12040757241	c_{61}	25
c_{14}	−4550993407	c_{30}	2944770693338	c_{46}	38731897791	c_{62}	−2
c_{15}	7091434955	c_{31}	10393055424026	c_{47}	−779482125		
c_{16}	22435828095	c_{32}	−4430027575756	c_{48}	−6967353269		

Theorem 3. *The recurrence above holds for every n , and it is the recurrence the entry records.*

Proof. It holds by Lemma 2, which returns a recurrence the sequence satisfies from its first term. For the identification: the entry asserts that the sequence satisfies SOME recurrence of order 62, possibly only from some index on. Let q be that recurrence’s characteristic polynomial and $q_{\min}$ the minimal polynomial of the tail of the sequence. Then $q_{\min} \mid q$. Berlekamp–Massey run on the sequence with its first $62 + 4$ terms removed returns order 62 as well, so $\deg q_{\min} = 62 = \deg q$; both are monic, so $q = q_{\min}$, which is the polynomial computed above. $\square$

4 Verification

Three checks.

First, the walk model reproduces all 17 terms the entry publishes, exactly, in integer arithmetic. This is what ties the model to the entry, and it is the only place where reading the entry’s English enters.

Second, the recovered recurrence was evaluated directly on those published terms, with no matrices and no Berlekamp–Massey involved, and holds at every index where the entry gives enough earlier terms to test it.

Third, the order was computed twice by independent routes: once modulo a 61-bit prime, where every intermediate is a machine word, and once in exact rational arithmetic; and once more on the sequence with its first $62 + 4$ terms removed, which is what the identification above requires. All three agree.

Remark 4. What is unusual here is that the conjecture’s statement was never read. The entry pins it down to a one-parameter family — the recurrences of order 62 that this sequence satisfies — and that family turns out to have exactly one member.

References

- [1] The OEIS Foundation, *The On-Line Encyclopedia of Integer Sequences*, <https://oeis.org>, 2026. Sequence A251266.
- [2] J. L. Massey, Shift-register synthesis and BCH decoding, *IEEE Trans. Inform. Theory* **15** (1969), 122–127.
- [3] R. P. Stanley, *Enumerative Combinatorics*, Volume 1, 2nd ed., Cambridge University Press, 2011. (Transfer-matrix method, Section 4.7.)
- [4] R. A. Horn and C. R. Johnson, *Matrix Analysis*, 2nd ed., Cambridge University Press, 2013.